Capacity-Law Serving of a 235B Mixture-of-Experts Model on 32 GB Host Memory

Routing Traces, Union Demand, and Why Stock Speculation Loses on a Consumer SSD

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Abstract

Qwen3-235B-A22B stores 235B parameters and activates about 22B per token [Yang et al., 2025]. In 4-bit GGUF form the checkpoint is 133 GB—larger than a 32 GB desktop's RAM. We serve it on one Windows/WSL2 workstation (RTX 3070 8 GB idle; CPU-only llama.cpp) by streaming MoE experts from NVMe with `O_DIRECT`. Random 3.5/10/16 MB `O_DIRECT` QD8 reads on the ext4 VHDX measure 2.34–2.37 GB/s. CPU decode with an 8-slot expert cache is 0.29 tok/s; a 12-slot cache that still fits in 21 Gi of WSL RAM reaches 0.38 tok/s (peak RSS 16.70 GiB). Full CPU mmap dies allocating a 105 GB `CPU_REPACK` buffer.

We release 94-layer top-8 routing traces (4×2048 tokens). Mean unique experts over a 4-token window is $U(4) = 18.98$, so mean unique demand exceeds 16 slots (naive k=4 still fits 23% of 16-slot windows). Identical lag-1 expert *sets* occur 0.26% of the time; an 8-slot admission cap has mean admitted length T=1.003. A 0.6B same-family draft is accepted at $\alpha = 0.75$–0.79 on a 16-token smoke, but stock llama.cpp verify with `-ub 1` *slows* that `llama-cli` 12-slot run to 0.80× (baseline 0.30 tok/s, not the 0.38 tok/s `llama-completion` greedy). Serial expert waves at 12 slots make multi-token verify legal without $3\,n_{\text{expert\_used}}$=24 resident slots; batched verify (`-ub 8`) returns wall-clock to $\approx 1.0\times$. Causal lag-2 prefetch recovers 0% of the LRU–OPT gap on these traces. A local `--moe-stream-umax` v1 ran on this box and *lost* (α=0, 0.79×/0.73×); we do not cite it as a HorizonSpec speedup. Trace-oracle $U_{\max}$ (Table 2) remains a simulation, not that engine result.

1 Introduction

Mixture-of-Experts (MoE) layers route each token to a top-k subset of experts [Shazeer et al., 2017, Fedus et al., 2022]. That sparsity is why a 235B-total / 22B-active model [Yang et al., 2025] can run at all on a workstation, but "runs" is not "interactive." Offloading Mixtral-8×7B with an LRU expert cache has been shown to reach 2–3 tok/s on a T4, RTX 3060, or RTX 3080 Mobile [Eliseev and Mazur, 2023]; MoE-Infinity reports 3.1–16.7× latency reduction from expert-level traces on a commodity GPU [Xue et al., 2024]. Those results do not transfer to a 133 GB Qwen3-235B table on 21 Gi of WSL RAM.

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This paper is a *measurement* study of that geometry. Direct I/O streaming of expert slabs is the substrate (llama.cpp PR#25294); we do not claim it as a novelty. Speculative decoding [Leviathan et al., 2023] supplies lookahead that greedy prefetch cannot. The systems question is whether the *union* of experts over a draft prefix fits in a RAM-sized slot budget—a *capacity law*, not a bandwidth slogan.

Contributions.

1. **A CPU tok/s ladder on one box** with provenance files: mmap impossible; 8-slot stream-direct 0.29 tok/s; 12-slot 0.38 tok/s at 16.70 GiB RSS.
2. **235B routing traces** of shape $(94, 2048, 8)$ on four domains, and the union law $U(k)$ that falsifies "just speculate with k=4."
3. **Paired speculation:** high α, negative wall-clock at `-ub 1`, parity after serial-wave batched verify.
4. **Negatives:** $U(4) \leq 16$ fails; causal prefetch recovery W=0; 8-slot HorizonSpec mean admitted length T=1.003.

A 0.44 tok/s figure from Windows `llama-server -ngl 99 -ncmoe 94` (GPU non-MoE trunk, 2026-08-05) is a *different stack* and is not used below. A companion training-time locality study is a separate artifact and is not mixed into these tok/s tables.

2 Background

MoE serving. At decode, each token's active experts must be present in a fast tier. EdgeMoE adapts expert bitwidth and keeps hot experts in an on-device buffer [Yi et al., 2023]. PowerInfer pins high-frequency neurons on GPU [Song et al., 2024]. Fiddler runs uncompressed Mixtral-8×7B by executing CPU-resident experts on the CPU [Kamahori et al., 2025]. FlexGen runs OPT-175B on a single 16 GB GPU at 1 tok/s generation *throughput* (effective batch size 144), not single-stream decode [Sheng et al., 2023]. Pre-gated MoE prefetches the next block's experts from the current block's gate [Hwang et al., 2023]. None of these papers report Qwen3-235B-A22B top-8 unions on a 12-slot, 21 Gi CPU stream cache.

Speculative decoding. A small draft proposes tokens; the target verifies a prefix in parallel with a sampling scheme that leaves the target distribution unchanged, giving 2×–3× wall-time acceleration on T5-XXL [Leviathan et al., 2023]. Medusa and EAGLE change how drafts are proposed, not the verify contract [Cai et al., 2024, Li et al., 2024]. On an MoE whose experts live on SSD, verify is cheap only if the *union* of experts over the prefix fits in the expert cache. That is the gap this paper measures.

Capacity law. Let s be the number of expert slots per MoE layer that fit in RAM, k the draft length, and $U(k)$ the number of unique experts in a k-token window at one layer. Greedy decode always loads $U(1)$ = top-8 = 8 experts per layer. A k-token verify is one I/O generation only if $U(k) \leq s$. If the engine instead instantiates a worst-case graph of $k \times 8$ experts, extra waves pay compute and I/O even when the realized union is small. Admission-time $U_{\max}$=s truncates the draft so the union fits. On traces we simulate longest-prefix admission with *target* routing (an upper bound). A local engine v1 (`--moe-stream-umax`) truncated the sampled+draft ubatch to

T=1 (log: admitting 1/3 and 1/5, uniq=8). That layout is expected for greedy-only union. Draft α also collapsed to 0: row-0 logits disagreed with F10b (not "T=1 dropped draft[0]"—the sampler still compares draft[0] at T=1). Empty-wave `skip_mul_mat_id` under plan_n_waves=1 is the leading unmeasured mechanism. The run is not a bit-exact A/B, not F4, and not a tok/s win. The F4 table is still per-layer-mean oracle routing.

3 Setup

Machine. Windows 11 + WSL2 Ubuntu, Intel i9-12900 (16C/24T), RTX 3070 8 GB idle, ~32 GB host RAM, `.wslconfig` 22 GB (WSL `free` ≈ 21 Gi), NVMe. llama.cpp branch `pr-25294`, git `1248fd8fa`, `GGML_CUDA=OFF`. Local patches on that commit: serial expert waves when $n_{\text{slots}} < 3\,n_{\text{expert\_used}} = 24$ (top-8, not draft length), ordered-wave waits (an OpenMP assert otherwise hangs), no next-wave pin when the cache is already full, and optional `--moe-stream-umax` (measured lose; not in Table 3). Model: `Qwen3-235B-A22B-Q4_K_M` (3-way split, 133 GB) and draft `Qwen3-0.6B-Q8_0` (768 MB, dense) on ext4. Expert cache `--moe-stream-direct --moe-stream-cache {8s,12s}`. Context 512, temperature 0, seed 1. Every headline number has a `config.json` / `verdict.json` under `week1/` (Appendix A).

I/O. fio 3.39, 20 s, 30B GGUF on the same VHDX (`week1/fio_summary.tsv`): mmap QD1 10 MB =0.064 GB/s; `pread` QD1 =1.377; `O_DIRECT` QD1 =1.856; `O_DIRECT` QD8 at 3.5/10/16 MB =2.368/2.339/2.354 GB/s.

30B check. Same binary, Qwen3-30B-A3B Q4_K_M: mmap 1.71 tok/s vs stream-direct 3.41 tok/s (1.99×). Direct I/O is not a no-op on this host.

4 The expert table versus the RAM slab

Qwen3-235B-A22B uses 94 MoE layers, 128 experts, top-8. At Q4_K_M an expert slab is 10.85 MB (`week1/uk_umax_235b.json`). Greedy decode therefore demands 94 × 8 × 10.85 MB = 7.97 GB of expert bytes per token before cache hits. A 12-slot cache is 12 × 94 × 10.85 MB = 11.96 GB of expert RAM; measured peak RSS is 16.70 GiB including activations and the dense trunk (`week1/ab235b_12s_verdict.json`). Sixteen slots would add ~4 GiB on top of 16.7 GiB and were not attempted. Full mmap failed: `ggml_aligned_malloc` of 105 496 MB for `CPU_REPACK`.

Feasibility is *expert-table bytes divided by RAM slab*, not parameter count. DeepSeek-V3 is 671B total with 37B activated per token [DeepSeek-AI, 2024]; a 4-bit table for that geometry is larger than this 133 GB Qwen3 checkpoint. We do not claim it is a smooth fit here.

5 Routing traces and the union law

`llama-moe-trace` records top-8 ids with weights untouched (stream-direct, 8-slot cache, 2048 tokens × 4 domains, window 512). Shape $(94, 2048, 8)$. Provenance: `/home/shri/week1/g0_235b/` and `week1/uk_235b.json`.

Union sharing is real ($U(4)/32 \approx 0.59$) but the union is larger than 16. Code $U(2) = 12.17$: mean unique experts over two tokens *exceeds* 12 slots. Naive k=2 fits 54% of 12-slot windows *per layer* (mean over layers), not a 4-token union and not engine min over 94 layers.

Table 1: Unique experts $U(k)$ and lag-1 set reuse on Qwen3-235B-A22B top-8 traces. $U(4) \leq 16$ fails on every domain. Source: `week1/uk_235b.json`.

domain	$U(4)$	$U(4)/32$	lag-1 reuse / chance	ratio
code	18.35	0.573	0.479 / 0.256	1.87×
general	18.87	0.590	0.463 / 0.203	2.27×
math	20.00	0.625	0.406 / 0.186	2.19×
medical	18.71	0.585	0.459 / 0.236	1.95×
mean	**18.98**	0.59	—	∼2.1×

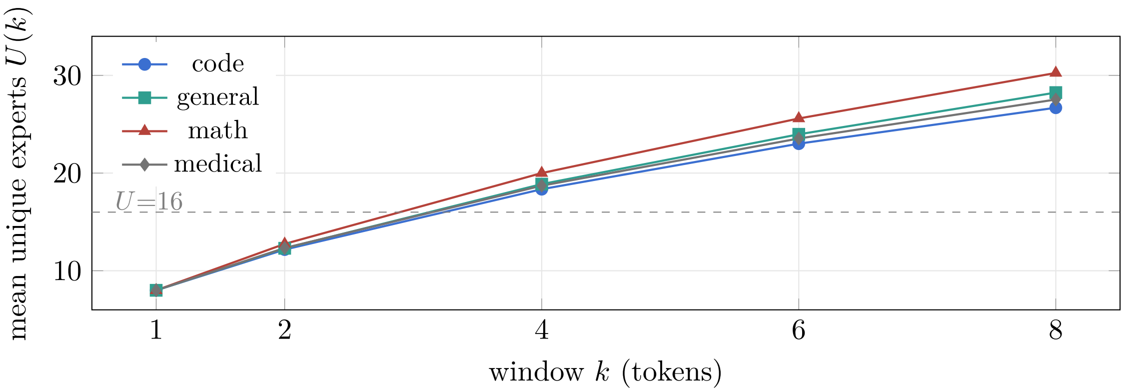

Figure 1: $U(k)$ grows through 16 by k=4 on every domain (`week1/uk_235b.json`). Mean $U(4)$ = 18.98 exceeds 16 slots; naive k=4 still fits 23% of 16-slot windows.

6 HorizonSpec is a slot count

Longest-prefix admission on these traces (target-oracle routing, an upper bound on union sharing) gives mean admitted length T. Consecutive tokens share an *identical* 8-expert set 0.26% of the time (`week1/uk_umax_235b.json` pooled). With top-8 routing, $U_{\max}$=8 almost never admits a second token. Mean T at 8 slots is 1.003—greedy.

At 12 slots, naive k=2 fits 54% of windows and HorizonSpec mean T is 1.54 *per layer*. The engine takes min admitted length over 94 layers (`week1/_min_layer_T.json`): pooled engine T=1.001, $p(T\geq 2)$=0.12%, ≈43 overflow layers per window. A global $U \leq 12$ gate therefore repeats T=1. Naive k=4 fits 1.2%: stock k=4 verify still wants ∼19 unique experts. That is why "just set draft k=4" is the wrong lever, and why 8-slot speculation is a no-op on this model independent of draft quality.

7 Decode and speculation

Greedy CPU decode (`llama-completion`, n=16, prompt "The capital of France is", `-b 1 -ub 1`). 8-slot page-cache 0.26 tok/s; 8-slot `O_DIRECT` 0.29 tok/s (`week1/ab235b_verdict.json`). 12-slot `O_DIRECT` (colder first arm) **0.38 tok/s**, RSS 16.70 GiB (`week1/ab235b_12s_verdict.json`). The same-day 8-slot rerun is warmer (0.30 tok/s) and is not averaged with the 12-slot number.

Draft quality. Qwen3-0.6B, `--spec-type draft-simple`. On 30B (not SSD-bound at 4 GiB cache) α = 0.857/0.833 at k=2/4 and wall-clock ≈ 1.0× (`week1/spec30b_verdict.json`, 37 gen tokens). On 235B, 16 eval tokens, Fibonacci prompt: α = 0.75 (k=2, 9/12) and 0.79 (k=4, 11/14).

Table 2: Target-oracle HorizonSpec on 235B traces. Per-layer mean, then mean over four domains. Not engine min over layers, not tok/s. Source: `week1/uk_umax_235b.json`.

cap	naive k=2 fit	HSpec k=2 T	naive k=4 fit	HSpec k=4 T
8 slots	0.26%	1.003	∼0%	1.003
12 slots	54.3%	1.54	1.2%	1.64
16 slots	100%	2.00	23.0%	2.83
24 slots	100%	2.00	94.6%	3.95

Table 3: 235B speculation, 12-slot stream-direct, n=16, seed 1. Eval tok/s from slot `print_timing` (`spec235b_verdict.json`, `spec235b_ub8/verdict.json`). Two-decimal tok/s; eval-ms/tok *speedup* vs this baseline is 0.951× / 0.977× (values <1 mean slower).

arm	`-ub 1` eval tok/s	`-ub 8` eval tok/s	`-ub 8` ms/tok	α
baseline	0.30	0.32	3089.56	—
spec k=2	0.24 (0.80×)	0.31 (0.97×)	3248.57	0.75
spec k=4	0.23 (0.77×)	0.32 (1.00×)	3162.69	0.79

Same-family drafting works. n=16 is a smoke, not a 200-prompt gate.

Wall-clock. llama.cpp with `-ub 1` serializes verify. At 12 slots, that *slows* 235B (Table 3). Stock overlap waves require $n_{\text{slots}} \geq 3\, n_{\text{expert\_used}}$=24 (Metal parking; the factor is top-8, not draft length k). We run *serial* waves with cap = n_{slots} when that inequality fails, wait if wave custom-ops enter out of order, and do not pin the next wave's residents when n_{slots} already equals plan capacity (that deadlock appeared on the second 4-token ubatch). Prefill `-ub 4` then completes at 0.28 prompt tok/s vs 0.16 at `-ub 1` on 16 tokens (`llama-completion`, `-b 1 -ub 1` then `-b 8 -ub 4`, second arm warmer; not the `llama-cli` 28-token prompt 0.44 tok/s).

Batched verify **stops the** 0.80× **tax**. It does **not** deliver the Table 2 *per-layer-mean* $T \approx 1.54$ projection (engine min T at 12 slots is 1.001): the graph is still sized for worst-case unique experts, Table 3 has no $U_{\max}$ (a later engine v1 admitted T=1 and lost α), and n=16 is a smoke. Skipping compiled empty-wave expert GEMMs (`week1/spec235b_skip/verdict.json`) leaves k=2 eval at 3251 ms/tok vs 3249 without the skip; k=4 is 3039 vs 3163 on a later, warmer arm—we do not claim a skip speedup. Peak RSS 16.7–17.6 GiB. `llama-cli -b 1` cannot greedy-decode this model (assert $n_{\text{tokens}} \leq n_{\text{batch}}$ after skipping the fused-GDN probe); the working invocation is `-b 512 -ub {1,8}`.

8 Prefetch is not the lever

On the same traces, a per-layer cache with prefetch *charging a slot*, objective = demand misses (`week1/g2a_235b.json`): at cap 16, lag-2 recovers 0% of the LRU→W8 gap; at caps 8 and 12 it *raises* demand misses. Frequency pinning is worse than LRU at cap 16 (code 307 vs 280 miss/tok). Non-causal oracle-next cuts demand misses ∼280→23. That ceiling is what a draft can supply only if verify can hold the union—the case for admission-time $U_{\max}$, not for lag-1 prefetch.

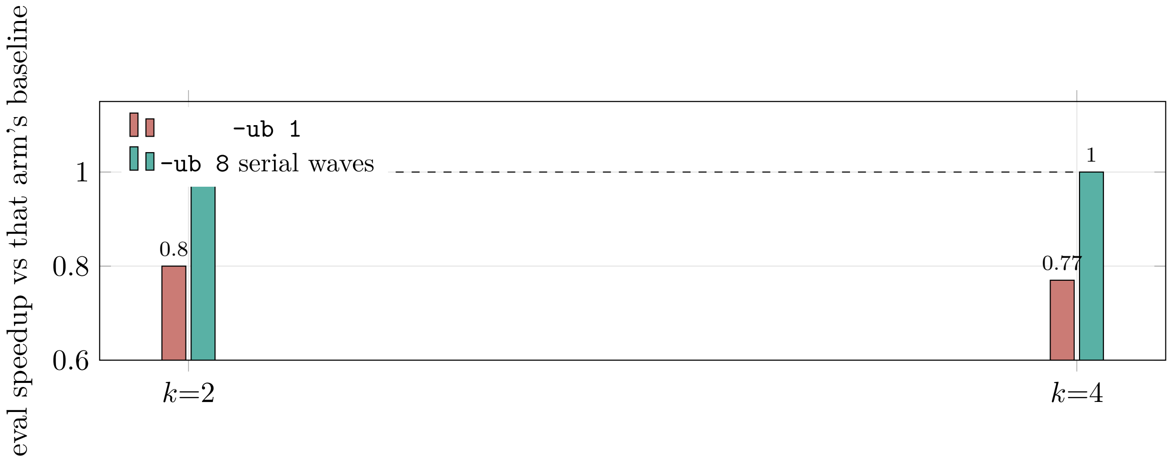

Figure 2: Stock `-ub 1` verify loses; serial-wave `-ub 8` returns near parity (2-decimal tok/s; k=4 eval-ms/tok *speedup* is 0.977×, i.e. slightly slower). Neither bar is a HorizonSpec engine speedup.

Table 4: Demand misses per token (sum over 94 layers) at 16-expert cap. Lag-2 equals LRU. Source: `week1/g2a_235b.json`.

domain	LRU	OPT W=8	lag-2	oracle-next	LRU hit
code	279.8	179.4	279.8	22.7	0.628
general	304.2	205.4	304.2	22.7	0.596
math	348.4	232.5	348.4	27.8	0.537
medical	293.6	193.2	293.6	25.1	0.610

9 Limitations

All 235B decode numbers are one prompt length and one seed unless stated. We did not run 200-prompt α. Table 2 is per-layer-mean oracle routing, not engine tok/s and not min over layers (that min is T=1.001 at 12 slots k=2; `_min_layer_T.json`). A `--moe-stream-umax` smoke admitted T=1 of a sampled+draft ubatch (α=0, 0.79×/0.73×; `spec235b_umax/verdict.json`); we do not treat prefix FFN as bit-exact—that was unmeasured. 12-slot serial waves are CPU-only; Metal still wants 24 slots. We did not load 235B on the 8 GB GPU. The 0.44 tok/s GPU-split figure is excluded. Empty-wave GEMM skip is a local tree change; Table 3 is the pre-skip `-ub 8` pair. The skip re-time does not move k=2 eval ms/tok.

10 Related work

Shazeer et al. and Switch established sparse expert routing [Shazeer et al., 2017, Fedus et al., 2022]. Edge and offload serving include EdgeMoE [Yi et al., 2023], PowerInfer [Song et al., 2024], Mixtral offload [Eliseev and Mazur, 2023], MoE-Infinity [Xue et al., 2024], Fiddler [Kamahori et al., 2025], and pre-gated prefetch [Hwang et al., 2023]. FlexGen studies offload scheduling [Sheng et al., 2023]. Speculative decoding verifies draft tokens in parallel [Leviathan et al., 2023, Cai et al., 2024, Li et al., 2024]. SiDA-MoE predicts experts for throughput [Du et al., 2024]. Concurrent MoE speculative-decoding papers also name the verify *union*: MoE-Spec loads a fixed expert budget B and drops the long tail [McDanel et al., 2026]; AcceptMoE restricts eligible experts and states that this changes the model distribution [Liang et al., 2026]; EcoSpec selects draft paths by predicted expert cost without changing the target verifier, on Qwen3-235B-A22B with 8 H200 GPUs [Xie et al., 2026]; EVICT truncates the draft tree before verify (same 235B model, tensor-parallel 8 on A100) [Pan et al., 2026]. Those are GPU/HBM or expert-*drop* levers. A 12-slot CPU stream cannot use a global $U \leq 12$ gate (engine min T=1.001); the remaining exact-routing lever is prefix FFN plus overflow waves. The closest measurement ancestors remain Mixtral offload (2–3 tok/s on 8×7B, not 235B) and MoE-Infinity (expert-level traces on a GPU that holds the working set). We contribute the 235B top-8 union law on a RAM slab that does *not* hold the table, and the speculation negative-then-parity pair under that law.

11 Conclusion

On this box, 235B Q4_K_M serving is a 12-slot, ≈0.3–0.4 tok/s CPU stream. Speculation has the right α and the wrong verify granularity until multi-token ubatch is legal; serial waves restore parity, not a 1.5× win. Per-layer-mean oracle $U_{\max}$ still says 54% of 12-slot k=2 windows fit (T=1.54); engine min over 94 layers is T=1.001 ($p(T \geq 2)$=0.12%). The engine v1 admitted one ubatch row and lost α. The remaining lever is: never skip row-0 FFN, always FFN the first draft row (serial waves if $U(2) > 12$), and do not set global `skip_mul_mat_id` under umax—not a global $U \leq 12$ gate. The publishable claims are the traces, the union law, the mmap/`O_DIRECT` ladder, and the speculation negative-then-parity pair—not umax tok/s. We post this as a measurement preprint. It is not a TMLR resubmission of the companion training-time locality paper.

Reproducibility. Scripts and verdicts live under `week1/`; traces under `g0_235b/`. llama.cpp `1248fd8fa` plus local serial-wave patches. CPU-only; GPU idle. Appendix A lists the file for each headline number.

Use of AI assistance. The author used an AI coding assistant to help implement the experimental harness and to draft and edit the manuscript. All experimental design decisions, the runs, and the interpretation are the author's, who takes full responsibility for the content and has verified every reported number against the listed artifacts.

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A Provenance

Table 5: Headline numbers and the files that contain them. Do not overlay rows that differ in binary, prompt, `n_ubatch`, or cache size.

claim	value	file under `week1/`
O_DIRECT QD8	2.34–2.37 GB/s	`fio_summary.tsv`
30B mmap / direct	1.71 / 3.41 tok/s	`ab30b/`
$U(4)$ mean	18.98	`uk_235b.json`
identical lag-1 sets	0.26%	`uk_umax_235b.json`
HSpec per-layer-mean T @ 12s k=2	1.54	`uk_umax_235b.json`
engine min T @ 12s k=2	1.001	`_min_layer_T.json`
mmap OOM	105 496 MB	`ab235b/mmap_repack_oom.stderr`
8s O_DIRECT greedy	0.29 tok/s	`ab235b_verdict.json`
12s O_DIRECT greedy	0.38 tok/s, 16.70 GiB	`ab235b_12s_verdict.json`
spec `-ub 1` k=2	0.24 tok/s, 0.80×	`spec235b_verdict.json`
spec `-ub 8` k=4	0.32 tok/s, 1.00× tok/s (0.977× ms/tok)	`spec235b_ub8/verdict.json`
empty-wave skip k=2	3251 vs 3249 ms/tok	`spec235b_skip/verdict.json`
umax v1 k=2	0.26 tok/s, α=0, 0.79×	`spec235b_umax/verdict.json`
lag-2 @ cap 16	0% of LRU–W8 gap	`g2a_235b.json`
30B spec α	0.86 / 0.83	`spec30b_verdict.json`